MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION

THIRD SEMESTER

MG3SDCREM201- Solar Thermal Technology(2025 ADMISSION ONWARDS)

Duration: 1.5hr **Maximum Marks:50**

Rememder(K) , ***Understand(U), Apply(A), Analyse(An),*** **Evaluate(E), Create(C), Skill(S),** ***Interest(I) and Appreciation(Ap)***

PART A

Short answer

Answer any 15 questions

Each question carries 1 marks

1. What is solar constant? [CO1-K]
2. Define diffuse radiation. [CO1-K]
3. Name the instrument used to measure direct solar radiation. [CO1-K]
4. What is a pyrometer? [CO1-K]
5. What is a sunshine recorder? [CO1-K]
6. What is a flat plate collector? [CO2-U]
7. Name the main types of flat plate collectors. [CO2-U]
8. What is a solar pond? [CO2-U]
9. What is a compound parabolic collector? [CO2-U]
10. What is a heliostat? [CO2-U]
11. What is a solar water heater? [CO4-An]
12. What is a solar cooker? [CO4-An]
13. What is a solar still? [CO4-An]
14. What is solar disinfection? [CO4-An]
15. What is solar greenhouse? [CO4-An]
16. What is direct gain in passive solar heating? [CO5-U]
17. What is radiation cooling? [CO5-U]

PART B

Short Essay

Answer any 3 questions

Each question carries 5 marks

18. Explain the formation of direct and diffuse solar radiation. [CO1-K]

19. Explain the construction and working of a liquid flat plate collector. [CO2-U]

20. Explain the construction and working of a two pass solar air heater [CO2-U]

21. Explain the construction and working of a simple solar still. [CO4-An]

22. Explain the components of an active solar heating system. [CO5-U]

PART C

Essay

Answer any two questions

Each question carries 10 marks

23. Explain the different types of solar dryers with their construction, working principles, advantages, disadvantages, and applications.

[CO4-An]

24. Explain the different passive solar heating techniques with their working principles and applications. [CO5-U]

25. Explain the measurement of solar radiation using a pyrometer, pyrheliometer, and sunshine recorder.

[CO1-K]

26. Explain the different types of flat plate air heating collectors with their construction and working principles. [CO2-U]